Name ______________ Date ________

LESSON 1

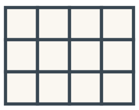
Volume with Whole-Number Cubes

SKILL 1

Find the volume of a right rectangular prism with whole-number edge lengths by packing it with unit cubes and counting.

Do Now

1 Consider an array of 3 rows of 4 squares. How many squares are there in all? Write a multiplication equation to show your thinking.

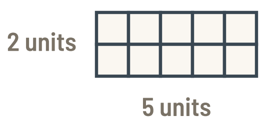


Equation: ______________________ There are ______ squares in all.

2 Solve.

3 × 4 = ____	4 × 2 = ____	2 × 5 = ____	5 × 6 = ____	4 × 7 = ____

3 Find the area of a rectangle that is 5 units by 2 units.



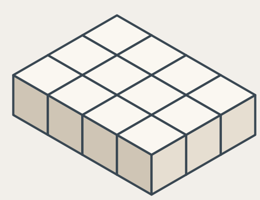
Area = ______ square units.

Vocabulary

TERM **Volume**

DEFINITION The amount of space a solid figure takes up, measured in cubic units.

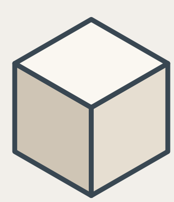
EXAMPLE A box packed with 12 one-inch unit cubes has a volume of 12 cubic units.



TERM **Unit Cube**

DEFINITION A cube that is 1 unit long, 1 unit wide, and 1 unit tall — the building block used to measure volume.

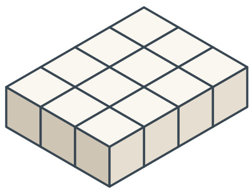
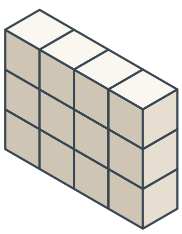
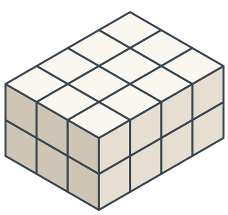
EXAMPLE A cube measuring 1 inch on every side is a unit cube. Many unit cubes packed together fill a larger box.



every edge = 1 unit

Guided Practice

Fill in the table as we work through the problems together.

Prism	Cubes per Layer	Number of Layers	Volume (cubic units)
1 			
2 			
3 			
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